BUSINESS REQUIREMENTS DOCUMENT

Citizen Self-Service Portal

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Problem

Northgate City Council processes ~180,000 citizen service requests per year across 5 departments — tracked in spreadsheets, routed manually, monitored by exception. Result: 38-day average resolution time, 58% SLA compliance, and £1.995M in avoidable annual operating cost.

Solution

End-to-end BA workstream delivering a PostgreSQL relational database (250K citizens · 1M service requests), 5 production SQL analytical queries, 3 Power BI dashboards, and a 25-KPI framework — built to GDS, UK GDPR, and WCAG 2.2 standards.

Impact

Projected 240% ROI over 3 years. Top recommendations target £1.1M in annual savings.

Fully synthetic dataset generated via PostgreSQL generate_series(). No real citizen data used at any stage. All organisations, names, and financial figures are fictional.

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1. Project Overview & Objectives

This Business Requirements Document (BRD) defines the functional requirements, data architecture, analytical queries, dashboard specifications, and delivery plan for the **Citizen Self-Service Portal** commissioned by Northgate City Council. The portal modernises public service delivery by replacing manual, spreadsheet-based request management with a fully integrated digital platform.

Baseline Performance (Current State)

KPI	Baseline	Target	Improvement
SLA Compliance %	58%	≥85%	+27pp
Average Resolution Time	38 days	≤30 days	–8 days
Citizen CSAT	3.1/5	≥4.0/5	+0.9 pts
Escalation Rate	22%	≤10%	–12pp
Cost per Transaction	£47	≤£28	–£19 (–40%)
Digital Adoption Rate	31%	≥75%	+44pp

Scope Summary

- 250,000 citizen profiles and 1,000,000 service requests in PostgreSQL 15
- 7 request types across 5 departments and 6 UK cities
- 5 production SQL analytical queries mapped to 3 Power BI dashboards
- 25-KPI framework with owners, targets, and reporting cadence
- Compliance with GDS Service Standard, UK GDPR, WCAG 2.2, Welsh Language Act
- 12-sprint Agile delivery plan across 4 phases (Q1–Q4 2025)

2. Database Schema

The data model operates under the **citizen_portal** PostgreSQL schema. Two core transactional/dimensional tables and one service-type lookup table support all analytical queries and Power BI dashboard calculations.

Table	Row Count	Description
Citizen_Service_Request	1,000,000	Core fact table — one row per service request
Citizen_Dim	250,000	Citizen profile dimension — linked via Citizen_ID
Service_Type_Dim	100	Lookup — service categories and SLA thresholds

Key Columns — Citizen_Service_Request

Column	Type	Notes
`Request_Type`	VARCHAR	Permit Renewal · Benefit Application · Tax Inquiry · Property Registration · License Application · Complaint · General Inquiry
`Department`	VARCHAR	Administration · Benefits · Revenue · Housing · Public Works
`Request_Priority`	VARCHAR	Critical · High · Medium · Low
`Request_Status`	VARCHAR	Submitted · In Review · Approved · Rejected · Completed · Escalated
`Service_Channel`	VARCHAR	Portal · Mobile App · Call Center · Walk-In
`City`	VARCHAR	London · Birmingham · Manchester · Leeds · Liverpool · Bristol
`Processing_Days`	INTEGER	0–120 — days from submission to resolution
`Satisfaction_Score`	INTEGER	1–5 (nullable — ~10% NULL across dataset)
`Escalation_Flag`	BOOLEAN	TRUE for ~8% of all requests
`Reopened_Count`	INTEGER	0–4 — number of times request was reopened
`Submission_Date`	DATE	Indexed — used in all time-series aggregations

SLA Threshold Definition

A **priority-aware SLA threshold** replaces the simple static 30-day cutoff used in legacy reporting. This variant is implemented as a CASE expression and should be built as a calculated column at load time so all queries and DAX measures share a single, consistent definition.

Priority	SLA Threshold	Rationale
Critical	≤3 days	Highest impact — citizen welfare, legal obligation
High	≤7 days	Significant impact — financial or housing matters
Medium	≤30 days	Standard service — majority of volume (~45%)
Low	≤60 days	Administrative — general inquiries and renewals

3. KPI Framework

The portal is governed by a 25-KPI framework. The six headline KPIs below drive executive dashboard reporting; full definitions, DAX formulas, data sources, and cadence are documented in **KPI-Definitions.xlsx**.

KPI	Baseline	Target	Owner	Cadence
SLA Compliance %	58%	≥85%	Dept Director	Weekly
Average Resolution Time	38 days	≤30 days	Ops Manager	Daily
Citizen CSAT	3.1/5	≥4.0/5	CX Lead	Monthly
Escalation Rate	22%	≤10%	Ops Manager	Weekly
Cost per Transaction	£47	≤£28	Finance	Monthly
Digital Adoption Rate	31%	≥75%	Digital Lead	Monthly

4. SQL Analytics Queries

Five production-ready queries, each mapped to a specific business question and Power BI dashboard visual. All queries target the **Citizen_Service_Request** table under the **citizen_portal** schema.

Query	Name	Key Columns	Business Question	Dashboard
Q-01	Monthly Request Volume Trend	`Submission_Date · Request_ID`	Where are the seasonal demand peaks?	Executive
Q-02	Top 5 Depts by SLA Breach Volume	`Department · Processing_Days · Request_Priority`	Which departments breach most, at which priority?	Exec + Ops
Q-03	Citizen Satisfaction by Channel	`Service_Channel · Satisfaction_Score`	Does channel measurably affect CSAT?	Citizen Exp.
Q-04	Escalation Rate — Rolling 3-Month Avg	`Submission_Date · Escalation_Flag`	Is escalation structurally rising or seasonal noise?	Operations
Q-05	Dept-wise Missing Satisfaction Scores	`Department · Satisfaction_Score`	Which dept CSAT data is statistically unreliable?	Citizen Exp.

Q-01 — Monthly Request Volume Trend

Attribute	Detail
Business Question	Where are the seasonal demand peaks, and is overall submission volume trending upward?
Key Columns	`Submission_Date · Request_ID`
Technique	TO_CHAR() for ISO month grouping · COUNT(*) aggregation by month
Dashboard	Executive Overview → Trend Analysis line chart (X = month, Y = total requests)

```
/* Q-01 · Monthly Request Volume Trend */ SELECT TO_CHAR(Submission_Date, 'YYYY-MM') AS month,
COUNT(*) AS total_requests FROM Citizen_Service_Request GROUP BY month ORDER BY month;
```

Findings

- *Submission volume peaks in Q1 (January–March) driven by Benefit Application and Tax Inquiry — together accounting for ~35% of all requests and concentrating in the first quarter.*
- *A secondary volume trough appears mid-year (July–August), creating a natural window for maintenance and backlog processing without service impact.*
- *Year-on-year volume growth is visible across the 2020–2026 dataset — current staffing models were not designed to absorb this trajectory.*

Q-02 — Top 5 Departments by SLA Breach Volume

Attribute	Detail
Business Question	Which departments breach SLA most, and does breach risk vary by Request_Priority?
Key Columns	`Department · Processing_Days · Request_Priority`
Technique	Static threshold (Processing_Days > 30) · COUNT(*) grouped by Department · ORDER BY DESC · LIMIT 5
Dashboard	Executive Overview → Department Ranking bar chart; Operations → Department Performance matrix

```
/* Q-02 · Top 5 Departments by SLA Breach Volume */ SELECT Department, COUNT(*) AS
sla_breach_count FROM Citizen_Service_Request WHERE Processing_Days > 30 GROUP BY Department
ORDER BY sla_breach_count DESC LIMIT 5;
```

Findings

- *Administration, Benefits, Housing, Revenue, and Public Works all appear in breach rankings — but their relative position shifts significantly when a priority-aware threshold replaces the static 30-day cutoff.*
- *Housing and Benefits generate the highest Critical and High priority breach counts — these cases carry the greatest citizen impact and highest penalty exposure.*
- *A request resolved in 25 days is compliant under a static 30-day SLA but a severe breach if it was flagged Critical (≤3 days) — the static threshold understates true compliance risk.*

Q-03 — Citizen Satisfaction by Channel (Digital vs Non-Digital)

Attribute	Detail
Business Question	Does the submission channel measurably affect Satisfaction_Score?
Key Columns	`Service_Channel · Satisfaction_Score`
Technique	CASE segmentation (Portal + Mobile App = Digital; Call Center + Walk-In = Non-Digital) · AVG(Satisfaction_Score) · normalised to 0–100% scale (AVG * 20)
Dashboard	Citizen Experience → Channel vs Satisfaction scatter / bar chart

```
/* Q-03 · Citizen Satisfaction by Channel (Digital vs Non-Digital) */ SELECT CASE WHEN
Service_Channel IN ('Portal', 'Mobile App') THEN 'Digital' ELSE 'Non-Digital' END AS
Channel_Type, COUNT(*) AS Total_Responses, ROUND(AVG(Satisfaction_Score), 2) AS
Avg_Satisfaction, ROUND(AVG(Satisfaction_Score) * 20, 2) AS Satisfaction_Percentage FROM
Citizen_Service_Request WHERE Satisfaction_Score IS NOT NULL GROUP BY 1;
```

Findings

- *Portal (~50% of all requests) and Mobile App (~25%) consistently return higher average Satisfaction_Score than Call Center (~15%) and Walk-In (~10%).*
- *The satisfaction gap between digital and non-digital channels widens for requests with Processing_Days above 14 — digital channels maintain scores during longer waits due to automated status visibility.*
- *Mobile App has grown substantially across 2020–2026 with no corresponding UX investment — the satisfaction advantage may erode without deliberate design maintenance.*

Q-04 — Escalation Rate Trend — Rolling 3-Month Average

Attribute	Detail
Business Question	Is the Escalation_Flag rate structurally rising, or are monthly spikes seasonal noise?
Key Columns	`Submission_Date · Escalation_Flag`
Technique	CTE aggregating monthly escalation rate · AVG() OVER window function (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) · raw vs smoothed rolling average
Dashboard	Operations → Escalation Trends combo chart (column = monthly raw, line = rolling 3-month average)

```
/* Q-04 · Escalation Rate Trend — Rolling 3-Month Average */ WITH monthly_escalation AS ( SELECT
DATE_TRUNC('month', Submission_Date) AS month, COUNT(*) AS total_requests, SUM(CASE WHEN
Escalation_Flag = TRUE THEN 1 ELSE 0 END) AS escalated_requests, ROUND( 100.0 * SUM(CASE WHEN
Escalation_Flag = TRUE THEN 1 ELSE 0 END) / COUNT(*), 2 ) AS escalation_rate_pct FROM
Citizen_Service_Request GROUP BY DATE_TRUNC('month', Submission_Date) ) SELECT month,
total_requests, escalated_requests, escalation_rate_pct, ROUND( AVG(escalation_rate_pct) OVER (
ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW ), 2 ) AS rolling_3_month_avg_pct FROM
monthly_escalation ORDER BY month;
```

Findings

- *Escalation_Flag = TRUE applies to approximately 8% of all 1M requests — but the rolling average reveals a structural upward trend, not a flat baseline with seasonal spikes.*
- *Monthly raw data shows Q1 spikes (consistent with the submission volume surge) — but the rolling average confirms these sit above an already-rising baseline, making Q1 a volume amplifier on top of a worsening underlying rate.*
- *Escalated cases concentrate in Housing and Benefits — the same departments with the highest Processing_Days and Reopened_Count averages, suggesting a compounding failure pattern.*

Q-05 — Department-wise Missing Satisfaction Scores Analysis

Attribute	Detail
Business Question	Which departments have CSAT data too incomplete to report reliably?
Key Columns	`Department · Satisfaction_Score`
Technique	COUNT(*) FILTER (WHERE Satisfaction_Score IS NULL) · null percentage per department · ranked DESC by missing rate
Dashboard	Citizen Experience → data quality callout; informs the [Satisfaction Rated Requests] DAX measure denominator

```
/* Q-05 · Department-wise Missing Satisfaction Scores Analysis */ SELECT Department, COUNT(*) AS
Total_Requests, COUNT(*) FILTER (WHERE Satisfaction_Score IS NULL) AS
Missing_Satisfaction_Scores, ROUND( 100.0 * COUNT(*) FILTER (WHERE Satisfaction_Score IS NULL) /
COUNT(*), 2 ) AS Missing_Percentage FROM Citizen_Service_Request GROUP BY Department ORDER BY
Missing_Percentage DESC;
```

Findings

- *Satisfaction_Score NULL rate averages ~10% across the dataset — but the null rate is not evenly distributed across departments.*

- *Walk-In and Call Center channels drive the highest null concentration — the agent closure workflow does not enforce score capture, unlike the Portal and Mobile App digital flows which prompt citizens directly.*
- *Departments exceeding a 15% null rate have CSAT averages based on a skewed subset of completions — publishing these figures as performance KPIs overstates or understates true satisfaction.*

5. Dashboard Specifications

Dashboard 1 — Executive Overview

Attribute	Detail
Audience	CIO, Directors, elected members
Visuals	KPI card strip · Monthly submission trend line · Department volume & SLA ranking · Geographic request distribution by city · Cross-page slicers (date, department, city, priority)

Key Findings

- *Housing department records the longest average Processing_Days and the highest SLA breach volume — a consistent pattern across all priority levels.*
- *Q1 submission volumes spike significantly above the annual monthly average — driven by Benefit Application and Tax Inquiry clustering in January and February.*
- *Benefit Application requests (~20% of volume) return the lowest Satisfaction_Score averages despite processing within SLA — an expectations and communications gap, not a delivery failure.*

Dashboard 2 — Operations

Attribute	Detail
Audience	Department heads, team managers, operations leads
Visuals	Stacked department bar by Request_Status · Priority breakdown donut · Escalation trend combo chart · Processing_Days distribution histogram · Department performance matrix

Key Findings

- *Escalated requests (~8% of all submissions) concentrated in Housing and Benefits, disproportionately in Critical and High priority cases.*
- *Medium priority requests (~45% of volume) — the largest single priority band — yet their average Processing_Days approaches the High priority SLA threshold in Housing.*
- *Reopened requests cluster in Benefits and Revenue, indicating first-contact resolution failures rather than capacity issues.*

Dashboard 3 — Citizen Experience

Attribute	Detail
Audience	CX leads, service designers, policy advisors
Visuals	Overall CSAT gauge (1–5 scale) · Satisfaction_Score distribution bar · Score by Request_Type · Channel usage treemap · Channel vs satisfaction scatter · Complaint volume by department and city

Key Findings

- *Portal and Mobile App (~75% of submissions combined) return higher average Satisfaction_Score than Call Center and Walk-In — digital-first citizens experience less friction.*

• Complaint requests (~5% of volume, smallest request type) show the longest average Processing_Days and the lowest satisfaction scores — the highest citizen attrition risk.

• Walk-In channel carries a disproportionately high Satisfaction_Score null rate — agents are not consistently recording scores at closure, making that channel's CSAT statistically unreliable.

6. Findings & Analytical Insights

Volume & Demand Patterns

- Submission volume peaks in Q1 each year — Benefit Application and Tax Inquiry together account for ~35% of all requests and cluster in January–February.
- A mid-year volume trough (July–August) creates a natural window for system maintenance and backlog processing.
- Year-on-year volume growth is visible across the 2020–2026 dataset; current staffing models were not designed to absorb this trajectory.

SLA Compliance

- The static 30-day SLA threshold used in legacy reporting systematically understates compliance risk — a request resolved in 25 days is compliant under the static rule but a severe breach for Critical priority (≤3 days).
- Housing and Benefits generate the highest Critical and High priority breach counts — the cases with the greatest citizen impact and highest penalty exposure.
- Administration, Benefits, Housing, Revenue, and Public Works all appear in breach rankings, but their relative positions shift materially when the priority-aware threshold is applied.

Escalation Patterns

- Escalation_Flag = TRUE applies to ~8% of all 1M requests. The rolling 3-month average reveals a structural upward trend, not seasonal noise.
- Q1 volume spikes amplify an already-rising escalation baseline — the Q1 spike is a volume multiplier, not the root cause.
- Escalated cases cluster in Housing and Benefits — the same departments with the highest Processing_Days and Reopened_Count, indicating a compounding failure pattern.

Citizen Satisfaction & Channel

- Portal and Mobile App channels (together ~75% of submissions) return higher average CSAT than Call Center and Walk-In — digital-first service meets citizen expectations more reliably.
- The CSAT gap between digital and non-digital channels widens for requests with Processing_Days above 14 — digital channels maintain satisfaction during longer waits via automated status updates.
- Complaint requests (~5% of volume) combine the longest Processing_Days of any request type with the lowest CSAT — the highest citizen attrition risk in the portfolio.
- Walk-In channel has a disproportionately high Satisfaction_Score null rate — agent closure workflows do not enforce score capture, making the channel's CSAT statistically unreliable.

Data Quality

- Satisfaction_Score is nullable with ~10% of rows set to NULL — but the null rate is not evenly distributed; Walk-In and Call Center drive the highest concentration.
- Departments exceeding a 15% null rate have CSAT averages based on a skewed subset — these figures should not be published as performance KPIs without a null-rate caveat.

7. Recommendations

The following five recommendations are derived directly from the SQL analytics findings. Each is quantified, owner-assigned, and prioritised using MoSCoW.

Replace Static 30-Day SLA Threshold	Must Have
Build the priority-aware CASE variant (Critical ≤3, High ≤7, Medium ≤30, Low ≤60) as a calculated column at database load time so every query and DAX measure uses a consistent definition. Legacy static threshold systematically understates compliance risk.	
Owner: **Data Lead** · Target: **Sprint 3**	

Enforce Satisfaction Score Capture on Assisted Channels	Must Have
Add a mandatory satisfaction score field (or explicit 'citizen declined' fallback) to the agent closure screen for Walk-In and Call Center channels. Digital channels already prompt citizens directly; assisted channels must reach parity to eliminate silent nulls.	
Owner: **Product Owner** · Target: **Sprint 5**	

Invest in Mobile App UX Maintenance	Should Have
The Mobile App channel has grown substantially (2020–2026) with no documented UX investment. The current satisfaction advantage over assisted channels will erode as volume scales if the experience is not actively maintained.	
Owner: **Digital Lead** · Target: **Sprint 8**	

Address the Benefits & Housing Compounding Failure Pattern	Should Have
Escalation, long processing times, and high reopened counts cluster in the same two departments. A combined 6-week workflow audit targeting Benefits and Housing simultaneously addresses multiple KPIs rather than treating each metric as an isolated problem.	
Owner: **Ops Manager** · Target: **Sprint 6**	

Build a Q1 Surge Staffing Protocol	Should Have
The January volume spike is visible and predictable in the dataset. Pre-authorising agency cover or overtime from the first working week of January each year would prevent the Q1 surge from converting into SLA breaches and escalations.	
Owner: **HR / Ops Manager** · Target: **Sprint 10**	

8. Delivery Phases & Agile Plan

Phase	Sprints	Focus	Target
1 — Foundation	1–3	Data model, citizen registration, submission form	Q1 2025
2 — Core Portal	4–6	Request tracking, notifications, SLA monitoring	Q2 2025
3 — Analytics	7–9	Power BI dashboards, KPI reporting, SQL analytics workstream	Q3 2025
4 — Enhancement	10–12	Mobile app, AI chatbot, demand forecasting	Q4 2025

The full Agile backlog comprises **8 epics** and **400+ story points** across 12 sprints. UAT is planned for 30 test cases covering functional, accessibility (WCAG 2.2), and performance scenarios. A 20-item risk register with RAG status and agreed mitigations is maintained in **Risk-Register.xlsx**.

9. UK Public Sector Compliance

Standard	Application
GDS Service Standard	User-centred design; accessible; multi-channel delivery
UK GDPR / DPA 2018	DPO embedded in stakeholder model; Privacy by Design; 7-year data retention
Equality Act 2010 / PSED	Vulnerable citizen fast-track SLA as a legal compliance obligation
WCAG 2.2 Level AA	Accessibility as a non-functional requirement and UAT test case
HM Treasury Green Book	Business case structured with NPV and 3-year ROI (240% projected)
Welsh Language Act 1993	Bilingual interface as FR-023; carried as a delivery risk in the risk register
CDDO Assisted Digital	Digital adoption risk mitigated via assisted digital programme

10. Tools & Techniques

Area	Detail
Database	PostgreSQL 15 — DDL, DML, CTEs, window functions, FILTER aggregation
Dashboarding	Power BI Desktop — DAX measures, relational model, cross-page slicers
Requirements	BRD, MoSCoW, functional and non-functional requirements, user stories with acceptance criteria
Stakeholder Management	Influence/interest mapping, RACI matrix, RAID log
Process Mapping	As-Is / To-Be swimlane diagrams
Agile Delivery	Scrum — epics, backlog, sprint planning, UAT, risk register
Business Case	HM Treasury Green Book format, ROI analysis, cost-benefit modelling
Compliance	UK GDPR, DPA 2018, WCAG 2.2, Privacy by Design

Document prepared by **Shahil Ameen, Business Analyst** · Northgate City Council Digital Transformation Programme · v1.0 — June 2025

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